Ex/EE/PC/B/T/223/2022

B.E. ELECTRICAL ENGINEERING SECOND YEAR SECOND SEMESTER EXAM 2022

Time: Three hours **Full Marks: 100**

POWER SUPPLY SYSTEMS

(50 marks for each part)

PART I

Solutions

Question 1

Problem Statement: Discuss about the following (i) surge tank of hydroelectric plant (ii) draft tube of hydroelectric plant (iii) reflector of nuclear power plant (iv) starting motor of gas turbine power plant (v) deaerator of steam power plant (vi) waste management in nuclear power plant.

Thought Process

Goal: Provide concise, technical definitions and functional descriptions for six key power plant components.
Methodology: Address each component individually, stating its primary purpose, location within the plant, and the operational problem it solves (e.g., draft tube recovers kinetic energy, deaerator removes corrosive gases).

Answer:

(i) **Surge Tank of Hydroelectric Plant:** A surge tank is an open-topped vertical standpipe or reservoir located near the turbine, connected to the penstock. Its primary function is to protect the penstock from the destructive high-pressure waves known as "water hammer." If the turbine suddenly rejects load and the governor closes the gates, the flowing water surges up into the tank, safely dissipating the kinetic energy. Conversely, it provides an immediate supply of water during sudden load increases.

(ii) **Draft Tube of Hydroelectric Plant:** A draft tube is a gradually expanding conduit that connects the exit of a reaction turbine (Francis or Kaplan) to the tailrace. It serves two main purposes: first, it allows the turbine to be installed above the tailrace level without losing any effective head. Second, its diverging shape converts the high kinetic energy of the exiting water into useful pressure energy, creating a negative pressure (suction) at the turbine runner exit, thereby increasing the overall efficiency of the turbine.

(iii) **Reflector of Nuclear Power Plant:** In a nuclear reactor, the core is surrounded by a reflector made of materials that scatter neutrons (like Graphite or Beryllium). During nuclear fission, fast neutrons tend to escape the core. The reflector bounces these escaping neutrons back into the core. This improves neutron economy, sustains the chain reaction more efficiently, and reduces the critical mass of the nuclear fuel required.

(iv) **Starting Motor of Gas Turbine Power Plant:** A gas turbine operating on the Brayton cycle is not self-starting. To initiate the cycle, the compressor must supply high-pressure air to the combustion chamber. The starting motor (usually an electric motor or small diesel engine) mechanically cranks the main turbine shaft to about 20% to 30% of its rated speed. Once the fuel is ignited and the turbine generates enough power to drive the compressor itself (self-sustaining speed), the starting motor automatically disengages via a clutch.

(v) **Deaerator of Steam Power Plant:** A deaerator is a specialized feed water heater used to remove dissolved corrosive gases, primarily oxygen and carbon dioxide, from the boiler feed water. It operates by heating the water with extraction steam to its saturation temperature, which reduces the solubility of these gases to near zero. Removing these gases is absolutely critical to prevent severe oxidation and pitting corrosion in the high-temperature, high-pressure boiler tubes.

(vi) **Waste Management in Nuclear Power Plant:** Nuclear waste is categorized by its radioactivity level. Low-level and intermediate-level wastes (contaminated clothing, tools, reactor resins) are usually compacted, solidified in concrete or bitumen, and stored in shielded steel drums in shallow geological repositories. High-level waste (spent nuclear fuel rods) generates intense heat and radiation. It is initially stored in deep cooling pools at the plant site for several years. Eventually, it undergoes reprocessing or is vitrified (encapsulated in glass) and sealed in heavily shielded casks for permanent disposal in deep, stable geological formations.

Question 2

Problem Statement (a): Discuss the feed water and steam flow circuit with respect to steam power plant with the help of a suitable schematic diagram.

Thought Process

Goal: Illustrate and explain the closed-loop Rankine cycle flow.
Methodology: 1. Use an orthogonal block diagram with properly spaced nodes to prevent word collapse. 2. Outline the path: Condenser → Feed Pump → Economizer → Boiler Drum → Superheater → Steam Turbine → Condenser. 3. Discuss the state of the fluid at each stage.

Answer (a):
Schematic Diagram:

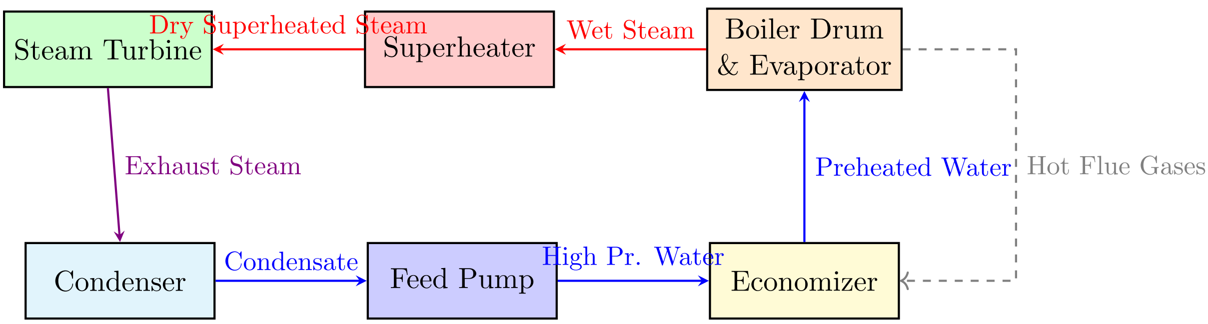

Figure 1: Feed Water and Steam Flow Circuit

Discussion of the Circuit:
The feed water and steam flow circuit is the primary thermodynamic loop of the Rankine cycle.

1. The cycle begins at the **Condenser**, where low-pressure exhaust steam from the turbine is cooled and condensed into liquid water (condensate).
2. The **Boiler Feed Pump** highly pressurizes this condensate to overcome the internal pressure of the boiler system.
3. The high-pressure water passes through the **Economizer**, where it is preheated by extracting residual heat from the outgoing hot flue gases, improving plant efficiency.
4. The preheated water enters the **Boiler Drum** and circulates through the furnace wall tubes. Heat from combustion converts the water into a saturated (wet) vapor.
5. The wet steam passes through the **Superheater**, where it absorbs more heat from the flue gases to become dry, high-temperature superheated steam, preventing moisture erosion in the turbine.
6. Finally, the superheated steam expands rapidly through the **Steam Turbine**, producing mechanical work to drive the alternator, before returning to the condenser.

Problem Statement (b): Explain the importance of using balanced draught over other artificial draught systems.
Answer (b):
A balanced draught system is a combination of both a Forced Draught (FD) fan, which pushes atmospheric air into the furnace, and an Induced Draught (ID) fan, which pulls flue gases out of the furnace. Its importance over singular artificial systems (using only FD or only ID) is profound in modern large-capacity boilers:

1. **Safety (vs. Pure Forced Draught):** A purely forced draught system operates at a positive pressure. This creates a severe safety hazard, as toxic flue gases, ash, and actual flames can leak outward into the boiler house through inspection doors and casing crevices. A balanced draught system precisely controls the fans to maintain a slight vacuum (negative pressure) inside the furnace, ensuring any leakage is inward (air leaking in) rather than outward.
2. **Efficiency (vs. Pure Induced Draught):** A purely induced draught system operates at a high negative pressure. This strong vacuum sucks excessive amounts of cold ambient air into the furnace through any cracks. This "tramp air" cools the furnace, reduces combustion efficiency, and overloads the ID fan. Balanced draught minimizes this negative pressure, optimizing the air-to-fuel ratio and maximizing thermal efficiency.
3. **Component Wear:** By sharing the workload, the massive pressure drops across the complex boiler passages (superheaters, economizers, air preheaters, ESPs) are handled efficiently without requiring a single, impractically large and heavily loaded fan.

Problem Statement (c): Draw a neat diagram of pressurized water nuclear reactor and describe the function of each component of such reactor.
Answer (c):
Diagram of a Pressurized Water Reactor (PWR):

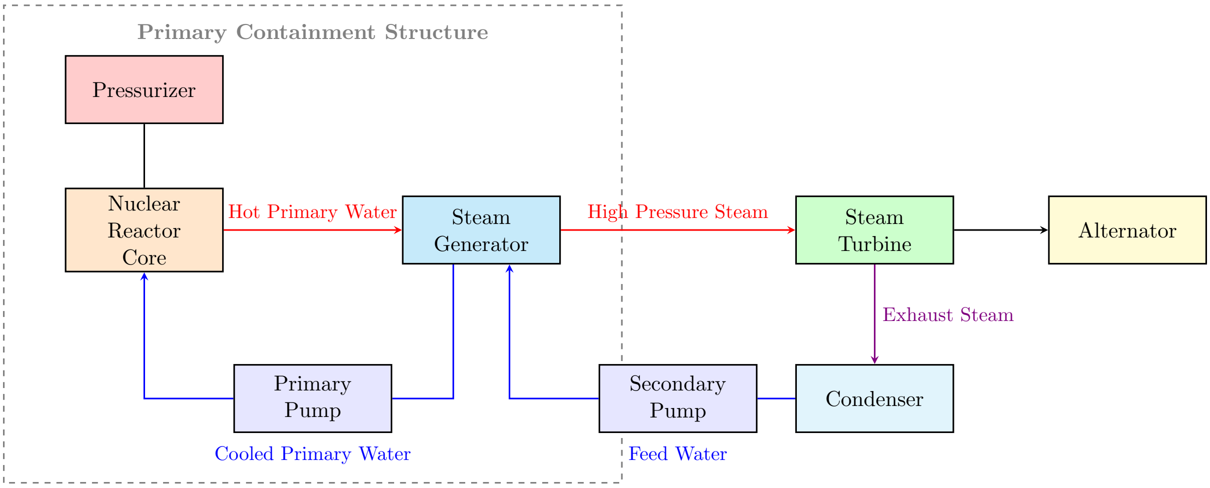

Figure 2: Schematic of a Pressurized Water Reactor (PWR)

Function of Components:
A PWR utilizes a two-loop design to keep radioactive primary coolant isolated from the secondary steam cycle.

1. **Reactor Core:** Contains the fissile nuclear fuel (U-235). It acts as the heat source, generating immense thermal energy via controlled nuclear fission.

2. **Pressurizer:** A tank connected to the primary loop. It maintains the primary coolant (light water) at an extremely high pressure (approx. 150 atm). This extreme pressure prevents the water from boiling, allowing it to remain liquid even at temperatures exceeding 300°C.

3. **Primary Pump:** Circulates the highly pressurized, radioactive light water through the reactor core (to act as both coolant and moderator) and pushes it to the steam generator.

4. **Steam Generator:** A massive heat exchanger. The hot, radioactive primary water flows through U-tubes. The non-radioactive secondary feed water flows around these tubes. Heat transfers across the metal, boiling the secondary water into steam.

5. **Secondary Loop (Turbine, Condenser, Pump):** Functions exactly like a conventional thermal plant. The generated clean steam drives the turbine, condenses, and is pumped back to the steam generator.

Question 3

Problem Statement (a): How many lubricating oil pumps used in steam power plant? Discuss about each of them in brief.

Answer (a):
In a modern high-capacity steam power plant (specifically for the turbo-alternator unit), there are typically **three** main lubricating oil pumps used to ensure the heavy turbine bearings are continuously supplied with oil under all operating conditions:

1. **Main Oil Pump (MOP):** This pump is mechanically coupled to and directly driven by the main turbine shaft. It is the primary pump during normal operation. Once the turbine reaches its rated synchronous speed, the MOP takes over the entire lubrication load, ensuring a continuous supply of oil proportional to the turbine speed.

2. **Auxiliary Oil Pump (AOP):** This is an AC electric motor-driven pump. Because the Main Oil Pump cannot supply sufficient oil pressure when the turbine is at a standstill or rotating slowly, the AOP is used during the start-up and shut-down phases of the plant. It provides the necessary lubrication until the turbine reaches a sufficient speed for the MOP to take over.

3. **Emergency Oil Pump (EOP):** This is a DC electric motor-driven pump, powered by the plant's backup battery banks. It is a critical safety device. If there is a total AC power failure (station blackout) while the turbine is still spinning down, the EOP automatically starts to supply lubricating oil. Without this, the heavy turbine shaft would rapidly destroy its bearings as it coasts down without lubrication.

Problem Statement (b): A 120 MW steam station uses coal of calorific values 6400 $kcal/kg$. Thermal efficiency of the station is 30% and electrical efficiency is 92%. Calculate the coal consumption per hr. when the station is delivering its rated full load.

Thought Process

Goal: Calculate hourly coal consumption at full load.
Methodology: 1. Determine output power: 120 MW = 120,000 kW. 2. Convert hourly electrical output to heat units: 1 kWh ≡ 860 kcal. 3. Find overall efficiency: $\eta_{overall} = \eta_{th} \times \eta_{elec}$. 4. Calculate required heat input: Output Heat / $\eta_{overall}$. 5. Calculate coal mass: Required Heat Input / Calorific Value.

Answer (b):
Given Data:

$$\begin{aligned} \text{Rated Power Output } (P) &= 120 \text{ MW} = 120,000 \text{ kW} \\ \text{Calorific Value of Coal } (CV) &= 6400 \text{ kcal/kg} \\ \text{Thermal Efficiency } (\eta_{th}) &= 30\% = 0.30 \\ \text{Electrical Efficiency } (\eta_{elec}) &= 92\% = 0.92 \end{aligned}$$

Step 1: Calculate the overall efficiency of the station.

$$\eta_{overall} = \eta_{th} \times \eta_{elec} = 0.30 \times 0.92 = 0.276 \text{ (or } 27.6\%)$$

Step 2: Calculate the electrical energy delivered in one hour.

$$\text{Energy Output per hour} = 120,000 \text{ kW} \times 1 \text{ hour} = 120,000 \text{ kWh}$$

Step 3: Convert the electrical energy output to its heat equivalent. Using the mechanical equivalent of heat (1 kWh = 860 kcal):

$$\text{Heat Equivalent of Output} = 120,000 \times 860 = 103,200,000 \text{ kcal/hr}$$

Step 4: Calculate the total heat input required from the coal.

$$\text{Total Heat Input} = \frac{\text{Heat Equivalent of Output}}{\eta_{overall}} = \frac{103,200,000}{0.276} = 373,913,043.48 \text{ kcal/hr}$$

Step 5: Calculate the coal consumption.

$$\begin{aligned} \text{Coal Consumption} &= \frac{\text{Total Heat Input}}{CV} \\ &= \frac{373,913,043.48 \text{ kcal/hr}}{6400 \text{ kcal/kg}} \\ &= 58,423.91 \text{ kg/hr} \end{aligned}$$

Result: The coal consumption is **58,423.91 kg/hr** (or roughly **58.42 metric tons per hour**).

Problem Statement (c): Discuss about air preheater used in steam power plant.

Answer (c):

The air preheater is a critical heat recovery component installed in the flue gas path of a steam power plant, typically situated after the economizer and before the electrostatic precipitator (ESP) or chimney.

- **Function:** Its primary function is to extract waste heat from the hot exiting flue gases and transfer it to the incoming cold atmospheric air (supplied by the Forced Draught fan) before this air is injected into the boiler furnace.
- **Benefits:**
    1. **Efficiency Improvement:** By recovering heat that would otherwise be lost up the chimney, it significantly increases the overall thermal efficiency of the plant. A rise of 20°C in air temperature can increase boiler efficiency by roughly 1%.
    2. **Combustion Stability:** Preheated air accelerates the ignition rate of the fuel, ensuring stable and complete combustion, which reduces soot formation.
    3. **Drying Pulverized Coal:** The hot air from the preheater is routed to the coal mills, where it dries the moisture out of the raw coal while simultaneously transporting the pulverized coal dust into the furnace.

Question 4

Problem Statement (a): What is hydrograph? Discuss different type of hydrograph also mention how does it help in designing and planning of hydro-electric power projects?

Answer (a):

What is a Hydrograph?

A hydrograph is a graphical representation showing the variation of water discharge (flow rate, typically in cubic meters per second, m^3/s) in a river or stream at a specific geographical measuring station, plotted against chronological time.

Different Types of Hydrographs:

Depending on the time scale chosen for the X-axis, hydrographs are classified as:

1. **Daily Hydrograph:** Plots the average daily flow over a month or season. Used for micro-level operational planning.
2. **Monthly Hydrograph:** Plots the average flow for each month over a year. It clearly shows the seasonal variations (e.g., wet monsoon months versus dry summer months).
3. **Annual Hydrograph:** Plots the average annual flow over several decades. Used to analyze long-term climate patterns and drought cycles.

Role in Designing and Planning:

Hydrographs are the absolute foundation of hydroelectric project planning:

- **Estimating Power Capacity:** The area under the hydrograph yields the total volume of water available. This determines the maximum average continuous power (firm power) the plant can generate.

- **Reservoir Sizing:** By comparing the hydrograph (natural flow) with the demand curve (water required for power generation), engineers determine the size of the dam and the reservoir capacity needed to store excess floodwater for use during dry seasons.
- **Spillway Design:** The peak points on a hydrograph reveal the maximum historically recorded flood rates. This data dictates the design capacity of the dam's spillways to prevent structural failure during extreme weather.

Problem Statement (b): Write a comparative study among pelton wheel, Francis turbine and Kaplan turbine.

Answer (b):

Parameter	**Pelton Wheel**	**Francis Turbine**	**Kaplan Turbine**
Type of Action	Pure Impulse	Reaction	Reaction
Water Head	High Head ($>$ 250 m)	Medium Head (30 to 250 m)	Low Head ($<$ 30 m)
Discharge Quantity	Low discharge	Medium discharge	High discharge
Flow Direction	Tangential flow	Mixed flow (Radial in, Axial out)	Axial flow
Specific Speed	Low (10 to 50)	Medium (50 to 250)	High (250 to 850)
Draft Tube	Not required	Required	Required (highly critical)
Runner Blades	Fixed bucket-shaped	Fixed curved vanes	Adjustable pitched blades

Problem Statement (c): What are the factors to be considered in site selection of hydroelectric power plant?

Answer (c):

The selection of a site for a hydroelectric power plant involves complex geological, geographical, and economic considerations:

1. **Water Availability:** The fundamental requirement is a continuous and reliable flow of water. Extensive historical hydrological data must verify sufficient yield.
2. **Water Head:** The site topography must provide a significant natural drop in elevation. A high head allows for smaller turbines and a smaller reservoir to generate the same power, reducing costs.
3. **Geology and Foundation:** The bedrock at the proposed dam site must be solid, stable, and free from faults or severe seismic activity to safely support the immense weight of the dam and the hydrostatic pressure of the reservoir without seepage.
4. **Topography for Reservoir:** The ideal site is a narrow gorge across the river (minimizing dam construction cost) that opens up into a wide, deep valley upstream (maximizing water storage capacity).

5. **Accessibility and Distance to Load Center:** The site must be accessible for transporting heavy machinery during construction. Additionally, if the site is too remote, the cost and line losses of transmitting the electricity to populated areas may render the project uneconomical.

Question 5

Problem Statement (a): What do you understand by the term 'turbo-alternator'?

Answer (a):

A **turbo-alternator** is a specialized type of synchronous AC generator (alternator) designed specifically to be coupled with high-speed prime movers, primarily steam turbines or gas turbines. Because these turbines are most efficient at very high rotational speeds (typically 3000 RPM for a 50 Hz system), the alternator must be robust enough to withstand immense centrifugal forces. Consequently, turbo-alternators are characterized by their smooth, non-salient pole (cylindrical) rotors made from solid steel forgings, featuring a relatively small diameter but a very long axial length.

Problem Statement (b): Differentiate between high head and low head hydroelectric plants.

Answer (b):

High Head Hydroelectric Plant	Low Head Hydroelectric Plant
Operates under water heads generally greater than 250 m.	Operates under water heads generally less than 30 m.
Requires a relatively small volume of water discharge to generate substantial power.	Requires a massive volume of continuous water discharge to generate power.
Utilizes Impulse turbines (primarily Pelton Wheels).	Utilizes Reaction turbines (primarily Kaplan turbines).
Involves long, high-pressure steel penstocks traversing down steep mountainous terrain.	Penstocks are very short or non-existent; water often flows directly from the barrage into the turbine casing.
Requires a large structural dam to artificially create the massive head.	Often built as run-of-river projects without massive dams, utilizing natural river flow or small barrages.

Problem Statement (c): With a schematic diagram, describe the principle of operation of a closed cycle gas turbine plant. Point out the merits of closed cycle gas turbine plant over open cycle gas turbine plant.

> Thought Process
>
> **Goal:** Detail the closed-cycle Brayton system and its advantages.
> **Methodology:** 1. Provide the standard orthogonal 4-component diagram (Compressor, Heater, Turbine, Cooler). 2. Explain the continuous circulation of the working fluid. 3. List merits focusing on fuel flexibility, cleanliness (no fouling), and thermal efficiency.

Answer (c):

Schematic Diagram:

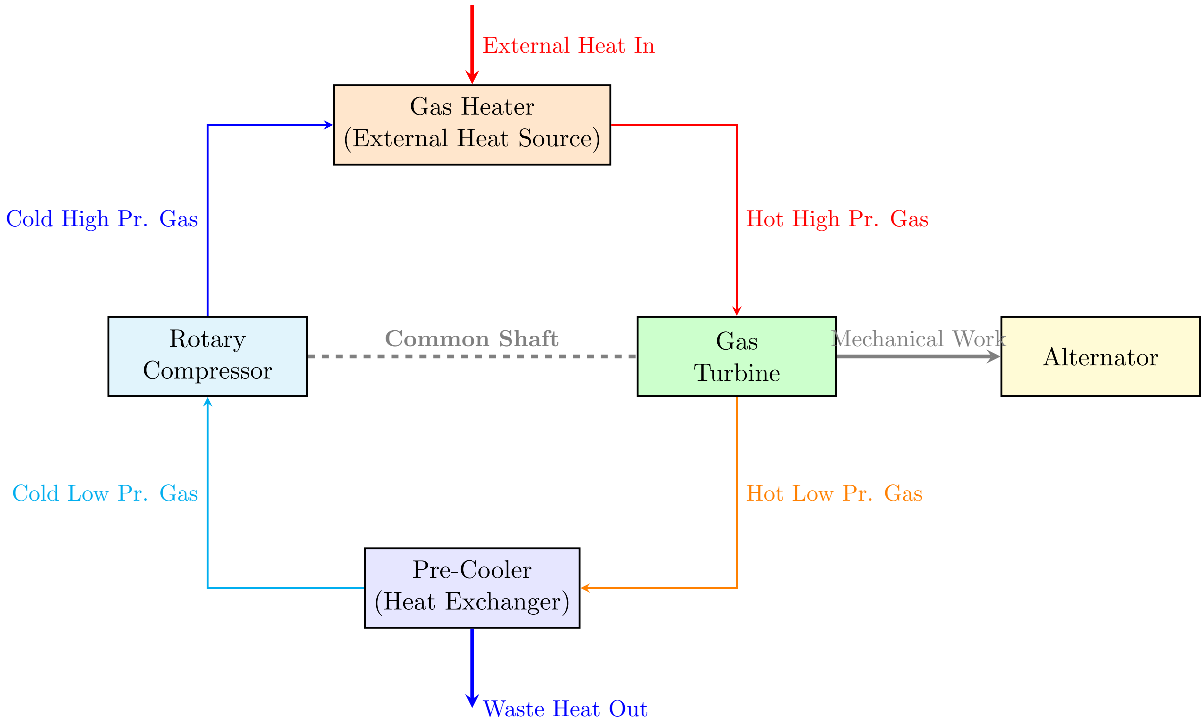

Figure 3: Principle of Operation of a Closed Cycle Gas Turbine Plant

Principle of Operation:

A closed-cycle gas turbine operates continuously on the constant-pressure Joule/Brayton cycle using a sealed working fluid (such as Helium, Argon, or dry air). The rotary compressor pressurizes the cold gas. This high-pressure gas passes through an external gas heater (heat exchanger), where thermal energy from a separate combustion furnace or nuclear reactor is transferred to the gas. The superheated, pressurized gas then expands through the gas turbine, generating mechanical work. Finally, the hot exhaust gas is routed through a pre-cooler to lower its temperature back to the initial state before it re-enters the compressor, repeating the cycle.

Merits Over Open Cycle Gas Turbine Plant:

1. **Fuel Flexibility:** Because heat is transferred externally via a heat exchanger, the combustion products never mix with the working fluid. Therefore, cheaper solid fuels like coal or peat can be used without damaging the turbine.

2. **No Turbine Fouling/Erosion:** The sealed working fluid remains completely pure. There are no abrasive ash particles or corrosive sulfur compounds passing through the turbine, drastically increasing blade lifespan and maintaining high efficiency over time.

3. **High Thermal Efficiency at All Loads:** The system can be operated at high baseline pressures throughout the loop. This results in superior heat transfer coefficients and allows the plant to maintain excellent thermal efficiency even when operating at partial loads by simply adjusting the mass flow of the fluid while maintaining operating temperatures.

4. **Smaller Size:** Operating at elevated internal pressures means the specific volume of the working fluid is smaller. Consequently, for the same power output, the physical dimensions of the compressor and turbine are significantly smaller compared to an open cycle system.